LAPORAN TUGAS 2: MIGRATION DAN SEEDER
Tabel Book dan Author - Sistem Informasi Penjualan Buku (Book Sales)

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1. Tujuan

Membuat tabel books dan authors menggunakan Laravel Migration sesuai desain database pada Brief, mengisi masing-masing 5 data dummy menggunakan Laravel Seeder, lalu menampilkannya melalui Model, Controller, Route, dan View.

2. Struktur Tabel (Migration)

Tabel	Kolom	Keterangan
authors	id, name, photo, bio, timestamps	name dan photo bertipe string, bio bertipe text
books	id, title, description, price, stock, cover_photo, genre_id, author_id, timestamps	price decimal(10,2), stock integer, author_id berelasi ke authors

Migration authors dibuat lebih dulu karena books memiliki foreign key author_id. Kolom genre_id disimpan sebagai angka biasa karena tabel genres tidak termasuk dalam tugas ini.

2.1 create_authors_table

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('authors', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->string('photo');
            $table->text('bio');
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('authors');
    }
};
```

2.2 create_books_table

```php
<?php
```

```
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('books', function (Blueprint $table) {
            $table->id();
            $table->string('title');
            $table->text('description');
            $table->decimal('price', 10, 2);
            $table->integer('stock');
            $table->string('cover_photo');
            // Tabel genres belum dibuat pada tugas ini, jadi genre_id disimpan sebagai angka biasa.
            $table->unsignedBigInteger('genre_id');
            $table->foreignId('author_id')->constrained('authors');
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('books');
    }
};
```

3. Data Dummy (Seeder)

Seeder dibuat untuk Author dan Book, masing-masing berisi 5 data. DatabaseSeeder memanggil AuthorSeeder lebih dulu, baru BookSeeder. Pada laporan ini hanya 2 data pertama yang ditampilkan.

3.1 AuthorSeeder

```
<?php

namespace Database\Seeders;

use App\Models\Author;
use Illuminate\Database\Seeder;

class AuthorSeeder extends Seeder
{
    public function run(): void
    {
        $authors = [
            ['name' => 'Andrea Hirata',         'photo' => 'andrea-hirata.jpg', 'bio' => 'Novelis Indonesia,
penulis tetralogi Laskar Pelangi.'],
            ['name' => 'Pramoedya Ananta Toer', 'photo' => 'pramoedya.jpg',     'bio' => 'Sastrawan
Indonesia, penulis Tetralogi Buru.'],
            // ... 3 data lainnya
        ];

        foreach ($authors as $author) {
            Author::create($author);
        }
    }
}
```

3.2 BookSeeder

```
<?php

namespace Database\Seeders;

use App\Models\Book;
```

```
use Illuminate\Database\Seeder;

class BookSeeder extends Seeder
{
    public function run(): void
    {
        $books = [
            ['title' => 'Laskar Pelangi', 'description' => 'Kisah sepuluh anak Belitung yang berjuang
bersekolah di tengah keterbatasan.', 'price' => 89000, 'stock' => 25, 'cover_photo' => 'laskar-pelangi.jpg',
'genre_id' => 1, 'author_id' => 1],
            ['title' => 'Bumi Manusia', 'description' => 'Novel sejarah tentang Minke di masa kolonial Hindia
Belanda.', 'price' => 120000, 'stock' => 15, 'cover_photo' => 'bumi-manusia.jpg', 'genre_id' => 5,
'author_id' => 2],
            // ... 3 data lainnya
        ];

        foreach ($books as $book) {
            Book::create($book);
        }
    }
}
```

4. Model, Controller, Route, dan View

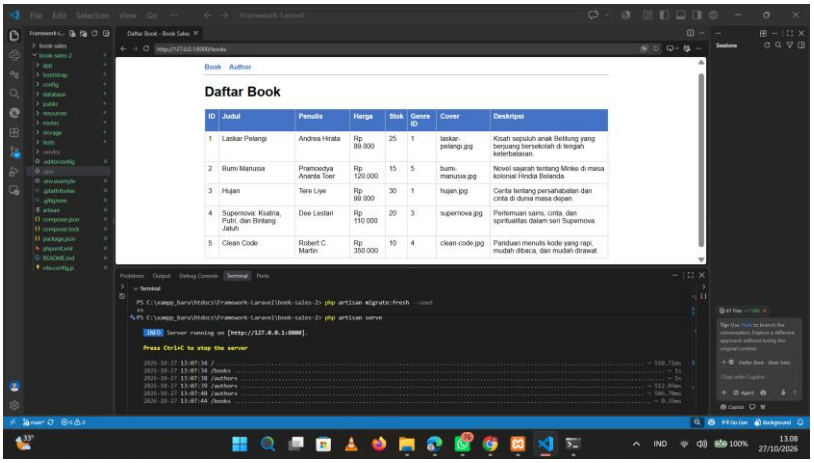
Komponen	Keterangan
Model	Author (hasMany books) dan Book (belongsTo author), dengan $fillable sesuai kolom tabel.
Controller	BookController@index mengambil data dengan Book::with("author"), AuthorController@index memakai Author::withCount("books").
Route	/books dan /authors, serta / yang diarahkan ke /books.
View	books/index.blade.php dan authors/index.blade.php, memakai layout layouts/app.blade.php.

5. Cara Menjalankan

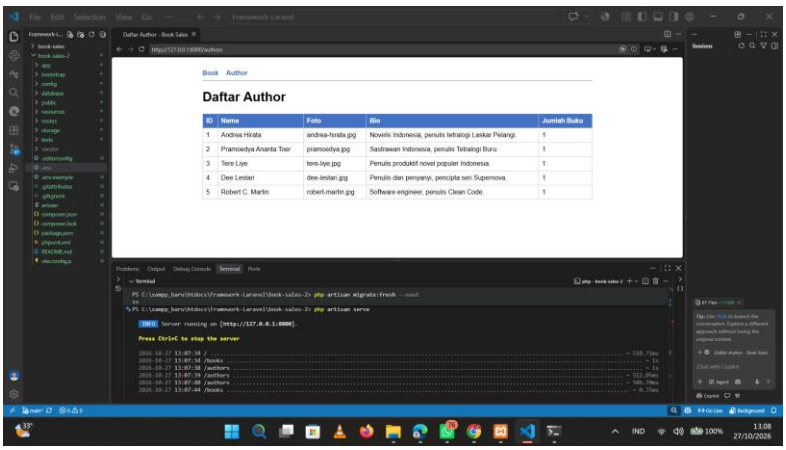
```
php artisan migrate:fresh --seed
php artisan serve
```

Perintah pertama membuat tabel dan mengisi data dummy. Perintah kedua menjalankan server, lalu halaman dibuka di http://127.0.0.1:8000/books dan /authors.

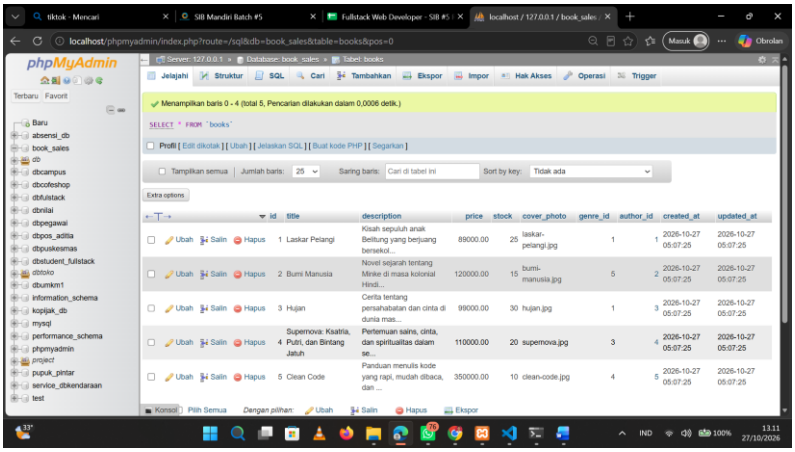
6. Hasil

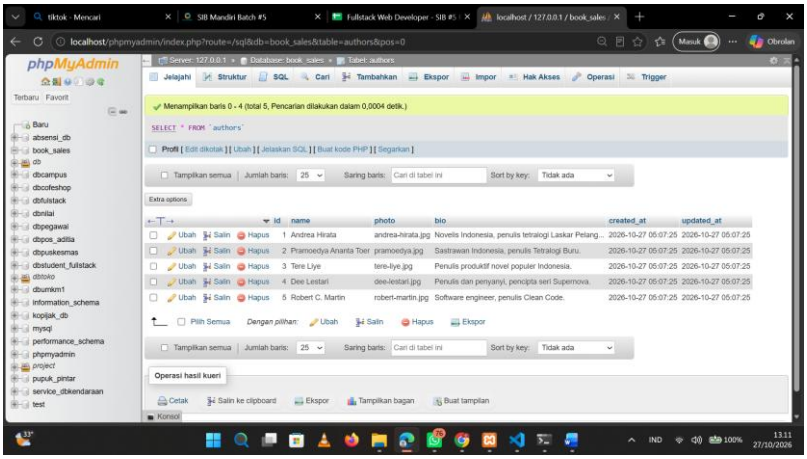


Gambar 1. Halaman /books menampilkan 5 data buku beserta nama penulis



Gambar 2. Halaman /authors menampilkan 5 data author beserta jumlah bukunya





Gambar 3. Tabel books dan authors pada database book_sales

7. GitHub

Link repository: https://github.com/muhammadaditia433/Framework-Laravel/tree/main/book-sales-2

File yang dikumpulkan: file migration (create_authors_table dan create_books_table) serta seeder (AuthorSeeder dan BookSeeder).